Fault-Tolerant Homomorphic Measurement for Lifted Product Codes

Research Progress

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Abstract

For CSS codes, homomorphic measurement provides a general framework for measuring logical Pauli operators by coupling the data block to an ancillary code through a suitable code homomorphism. However, only a limited number of code families currently have both fault-tolerant ancilla constructions and explicit homomorphisms that support addressable logical measurements. In this work, we investigate fault-tolerant constructions for addressable homomorphic measurement in lifted product codes. Our goal is to identify ancillary lifted product codes, compatible homomorphisms, and fault-tolerant preparation and readout procedures that implement selected logical measurements while preserving the remaining logical information.

1 Background

1.1 CSS codes as three-term chain complexes

A CSS code is represented by a three-term chain complex over $\mathbb{F}_2$,

$$Q: \qquad Q_2 \xrightarrow{\partial_2^Q} Q_1 \xrightarrow{\partial_1^Q} Q_0, \qquad \partial_1^Q \partial_2^Q = 0. \tag{1}$$

The bases of Q_2, Q_1, and Q_0 index the Z checks, qubits, and X checks, respectively. Thus

$$H_X = \partial_1^Q, \qquad H_Z^T = \partial_2^Q, \qquad H_X H_Z^T = 0. \tag{2}$$

Logical Z operators are homology classes and logical X operators are cohomology classes,

$$H_1(Q) = \frac{\ker \partial_1^Q}{\operatorname{im} \partial_2^Q}, \qquad H^1(Q) = \frac{\ker (\partial_2^Q)^T}{\operatorname{im} (\partial_1^Q)^T}. \tag{3}$$

1.2 Lifted-product codes

1.2.1 The ring-valued product complex

Let

$$R = \mathbb{F}_2[x]/(x^\ell + 1). \tag{4}$$

Over $\mathbb{F}_2$, $x^\ell + 1 = x^\ell - 1$, so multiplication by x is the cyclic shift of a length-ℓ binary fibre. For $f \in R$, write $\overline{f} = f(x^{-1}) = f(x^{\ell-1})$, and for a matrix C over R define $C^* = \overline{C}^T$. Consider two one-term complexes of free R-modules,

$$A_\bullet: \ R^{n_A} \xrightarrow{A} R^{m_A}, \qquad B_\bullet: \ R^{n_B} \xrightarrow{B} R^{m_B}, \tag{5}$$

where $A \in R^{m_A \times n_A}$ and $B \in R^{m_B \times n_B}$. The two-dimensional lifted product $\mathrm{LP}_\ell(A, B)$ is the total complex $M = A_\bullet \otimes_R B_\bullet$,

$$M_2 \xrightarrow{\partial_2^M} M_1 \xrightarrow{\partial_1^M} M_0, \tag{6}$$

with

$$\begin{aligned} M_2 &= R^{n_A} \otimes_R R^{n_B}, \\ M_1 &= (R^{n_A} \otimes_R R^{m_B}) \oplus (R^{m_A} \otimes_R R^{n_B}), \\ M_0 &= R^{m_A} \otimes_R R^{m_B}, \end{aligned} \tag{7}$$

and boundary maps

$$(\partial_2^M)^T = (I_{n_A} \otimes_R B^*,\ A^* \otimes_R I_{n_B}), \qquad \partial_1^M = (A \otimes_R I_{m_B},\ I_{m_A} \otimes_R B). \tag{8}$$

These are the conventions of Definition III.1 in Ref. [1]. In particular, $\partial_1^M \partial_2^M = 0$.

1.2.2 Binarization and the CSS code

The monomial basis $1, x, \ldots, x^{\ell-1}$ defines an $\mathbb{F}_2$-linear isomorphism

$$\mathsf{B} : R^s \longrightarrow \mathbb{F}_2^{s\ell}. \tag{9}$$

It also gives the regular binary representation of an R-linear matrix and satisfies $\mathsf{B}(Cu) = \mathsf{B}(C)\mathsf{B}(u)$. Applying B to Eq. (6) gives a binary chain complex $Q = \mathsf{B}(M)$ and hence a CSS code with

$$H_X = \mathsf{B}(\partial_1^M), \qquad H_Z^T = \mathsf{B}(\partial_2^M). \tag{10}$$

The physical-qubit count is therefore

$$n = \ell(n_A m_B + m_A n_B). \tag{11}$$

The map B is a bijective chain map, so it identifies the ring-valued homology of M with the binary logical-Z space of Q. When ℓ is odd, R is semisimple and the Künneth decomposition takes the schematic form

$$H_1(M) \simeq (\ker_R A \otimes_R \operatorname{coker}_R B) \oplus (\operatorname{coker}_R A \otimes_R \ker_R B). \tag{12}$$

This decomposition explains the two physical sectors in M_1 and is the starting point for constructing explicit logical bases.

1.2.3 Canonical logical fibres

Zheng *et al.* identify a canonical sector when compatible information sets can be chosen for the relevant kernel and cokernel modules. Put $r_A = n_A - m_A$ and $r_B^* = m_B - n_B$. Its logical qubits are indexed by

$$(i, j, m) \in [r_A] \times [r_B^*] \times \mathbb{Z}_\ell, \qquad k_{\mathrm{c}} = r_A r_B^* \ell. \tag{13}$$

For suitable systematic representatives $u_{A_1}^{(i)}, e_{A_1}^{(i)}, e_{B_0}^{(j)}, v_{B_0}^{(j)}$, a conjugate logical pair has ring-valued supports

$$\operatorname{Supp} \overline{Z}_{i,j,m} = x^m u_{A_1}^{(i)} \otimes_R e_{B_0}^{(j)}, \tag{14}$$

$$\operatorname{Supp} \overline{X}_{i,j,m} = x^m e_{A_1}^{(i)} \otimes_R v_{B_0}^{(j)}. \tag{15}$$

Their binary supports are obtained with B. The Z and X representatives intersect on exactly one physical qubit precisely when all three indices agree. Multiplication by x^t shifts the third index, $m \mapsto m + t \pmod{\ell}$. Thus an HGP-like row–column grid is replaced by a grid of length-ℓ logical fibres. The canonical sector need not exhaust the full logical space; any residual logical sector must be tracked separately when claiming addressability or nondestructive action on spectator qubits.

1.3 Homomorphic CNOT and homomorphic measurement

1.3.1 Chain maps as physical CNOT patterns

Let Q be a data code and Q' an ancilla code, both written as in Eq. (1). A chain map $\gamma : Q' \to Q$ is a triple

$$\gamma_i : Q'_i \longrightarrow Q_i, \qquad i = 0, 1, 2, \tag{16}$$

satisfying

$$\partial_i^Q \gamma_i = \gamma_{i-1} \partial_i^{Q'}, \qquad i = 1, 2. \tag{17}$$

The map γ_1 specifies the physical CNOTs: data qubit i controls ancilla qubit j whenever $\gamma_1[i, j] = 1$. Equation (17) ensures stabilizer preservation. Coordinate inclusions are transversal, while bounded-degree maps have bounded CNOT depth.

The logical action is most transparent on (co)homology. The chain map induces

$$\gamma_* : H_1(Q') \longrightarrow H_1(Q), \qquad \gamma^* : H^1(Q) \longrightarrow H^1(Q'). \tag{18}$$

For a data-control, ancilla-target CNOT, an ancilla logical-Z class $[z']$ and a data logical-X class $[x]$ transform schematically as

$$\overline{Z}_{Q'}([z']) \longmapsto \overline{Z}_Q(\gamma_*[z'])\, \overline{Z}_{Q'}([z']), \tag{19}$$

$$\overline{X}_Q([x]) \longmapsto \overline{X}_Q([x])\, \overline{X}_{Q'}(\gamma^*[x]). \tag{20}$$

The two induced maps therefore have to be checked together: specifying the desired image of a Z logical does not by itself establish the full logical Clifford action.

1.3.2 Generalized Steane measurement

The homomorphic CNOT becomes a nondestructive logical measurement through a generalized Steane circuit. For a Z-type measurement, prepare the relevant logical qubits of Q' in the logical Z basis, apply the data-control homomorphic CNOT, and measure Q' in the physical Z basis. Each decoded ancilla logical outcome measures the data logical product represented by the corresponding class in $\operatorname{im} \gamma_*$. For an X-type measurement, prepare the ancilla in the logical X basis, reverse the physical CNOT direction, and measure the ancilla in the physical X basis. Multiple ancilla logical qubits give multiple commuting Pauli-product measurements in parallel.

A chain map proves that the ideal CNOT pattern preserves the codespace, but it is not by itself a fault-tolerance proof. The construction in Ref. [2] additionally uses an ancilla whose distance is at least the data distance, fault-tolerant logical-state preparation, a constant-depth or transversal coupling, repeated syndrome extraction, transversal readout, and classical decoding or Pauli-frame processing. These are separate obligations in an LP extension.

1.4 HGP homomorphic CNOTs

1.4.1 HGP codes inside the LP convention

Let $H_a \in \mathbb{F}_2^{r_a \times n_a}$ be full-row-rank classical check matrices, with $k_a = n_a - r_a$ for $a = 1, 2$. Xu *et al.* use H_1^T as the vertical chain map and H_2 as the horizontal chain map. In the LP notation above, this is the specialization

$$\ell = 1, \qquad A = H_1^T, \qquad B = H_2. \tag{21}$$

After swapping the two direct summands of M_1, the resulting HGP check matrices are

$$H_X = (I_{n_1} \otimes H_2,\ H_1^T \otimes I_{r_2}), \tag{22}$$

$$H_Z = (H_1 \otimes I_{n_2},\ I_{r_1} \otimes H_2^T). \tag{23}$$

The code has parameters

$$[[n_1 n_2 + r_1 r_2,\ k_1 k_2,\ \min(d_1, d_2)]], \tag{24}$$

under the full-rank assumptions used in Ref. [2].

Choose systematic kernel generators $b_i^{(1)} \in \ker H_1$ and $b_j^{(2)} \in \ker H_2$, together with unit vectors on the corresponding information coordinates. A canonical logical grid is

$$\overline{Q} = \{\overline{Q}_{i,j}\}_{i\in[k_1],j\in[k_2]}, \tag{25}$$

with representatives, in the bit–bit qubit sector,

$$\overline{X}_{i,j} = \begin{pmatrix} b_i^{(1)} \otimes e_j^{(n_2)} \\ 0 \end{pmatrix}, \qquad \overline{Z}_{i,j} = \begin{pmatrix} e_i^{(n_1)} \otimes b_j^{(2)} \\ 0 \end{pmatrix}. \tag{26}$$

The X representative occupies a logical column, the Z representative a logical row, and conjugate representatives meet at one information-coordinate qubit.

1.4.2 Puncturing, augmenting, and the induced CNOT

Consider a collection of disjoint horizontal hyperedges $E_h = \{e_\alpha\}$ with $e_\alpha \subseteq [k_2]$. Let

$$S = [k_2] \setminus \bigcup_{e\in E_h} e \tag{27}$$

be the unused horizontal information coordinates. Starting from H_2, add repetition-code checks $H_{\mathrm{rep}}^{E_h}$ that identify the logical bits inside each e_α, and then puncture the coordinates in S:

$$H_2' = (H_2^{+H_{\mathrm{rep}}^{E_h}})^S, \qquad Q' = \mathrm{HGP}(H_1, H_2'). \tag{28}$$

Augmentation expurgates unwanted codewords, while puncturing information coordinates removes unused logical columns. The coordinate inclusion from H_2' to H_2, tensored with the identity on the vertical factor, induces a chain map $\gamma : Q' \to Q$. Its qubit component identifies the shared HGP coordinates, so the physical homomorphic CNOT is transversal on those coordinates.

For each row i and hyperedge e_α, one logical qubit of Q' couples to all data logical qubits $\{\overline{Q}_{i,j} : j \in e_\alpha\}$. Up to stabilizers, the data-control, ancilla-target CNOT acts as

$$\overline{Z}'_{i,\alpha} \longmapsto \overline{Z}'_{i,\alpha} \prod_{j\in e_\alpha} \overline{Z}_{i,j}, \tag{29}$$

$$\overline{X}_{i,j} \longmapsto \overline{X}_{i,j}\overline{X}'_{i,\alpha}, \qquad j \in e_\alpha, \tag{30}$$

while the remaining logical operators are unchanged. Preparing Q' in the logical Z basis and measuring it after this CNOT therefore measures

$$\left\{ \prod_{j\in e} \overline{Z}_{i,j} : i \in [k_1],\ e \in E_h \right\} \tag{31}$$

in parallel. This is the horizontal-PPM gadget of Algorithm 1 and Theorem 9 in Ref. [2].

The fault-tolerance argument uses two special properties of this construction. First, the punctured coordinates are information bits, and the added checks only expurgate codewords; consequently the modified horizontal classical distance does not decrease. Second, Q' is prepared by transversal physical $|0\rangle$ preparation followed by $d = \min(d_1, d_2)$ rounds of stabilizer measurement, coupled transversally to Q, and read out transversally. Thus the ancilla distance is at least the data-code distance under the assumptions of the theorem.

1.4.3 From row-parallel to addressable grid measurements

The horizontal construction selects logical columns or column products, but it repeats the same measurement in every logical row. Xu *et al.* obtain a selected grid pattern by adding a second HGP block Q'', called the mask code. A vertical puncturing–augmenting homomorphism prepares selected logical qubits of Q' in $|+\rangle$, or prepares groups of them in logical GHZ states, before the final horizontal-Z measurement. Because the vertical factor is represented by a transpose, the vertical homomorphism and its physical CNOT reverse direction. Exchanging X and Z gives the dual vertical-X construction. Combining the two stages yields grid Pauli-product measurements on products of disjoint row and column hyperedges.

1.5 Requirements for an LP extension

Equations (14)–(15) suggest an LP analogue of the HGP logical grid, but the extra fibre index m creates additional addressability requirements. A fault-tolerant construction should establish all of the following rather than only an intended action on one family of logical Z operators:

1. construct explicit R-linear ancillary complexes and chain maps, and identify their binary CNOT schedules after applying B;
2. compute both induced maps γ_* and γ^* on the entire logical space, including any residual sector outside the canonical fibres;
3. select the desired (i, j, m) logical qubits while proving that unmeasured canonical and residual logical information is preserved;
4. bound the ancilla distance and the distance of the coupled or merged code, rather than inferring fault tolerance from the chain-map identity;
5. give fault-tolerant ancilla preparation, repeated syndrome extraction, readout, decoding, and Pauli-frame procedures; and
6. verify that $\mathsf{B}(\gamma_1)$ has bounded degree and admits a constant-depth or otherwise quantified physical CNOT schedule.

The HGP construction supplies a concrete template—modify base complexes, derive the chain map, calculate its complete logical action, and only then add the preparation and readout protocol. The next step is to determine which of these operations remain internal to the LP family over R and which require a separate mask or extractor construction.

References

[1] H. Zheng, G. Zheng, L. Jiang, and Q. Xu, "Logical Computation with Canonical Lifted Product Codes," arXiv:2607.28605v1 (2026).

[2] Q. Xu, H. Zhou, G. Zheng, D. Bluvstein, J. P. Bonilla Ataides, M. D. Lukin, and L. Jiang, "Fast and Parallelizable Logical Computation with Homological Product Codes," arXiv:2407.18490v1 (2024).